

LittleToDo: Large Language Model Driven Intervention Tool for Adolescent Academic Procrastination with Affective Computing

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Abstract

This paper explores the causes and intervention methods of adolescent academic procrastination and proposes LittleToDo, a compact on-desk device designed to support after-school learning tasks completion by reduce procrastination behaviors. LittleToDo integrates an AI agent as its core component and utilizes a miniature printer as its interaction medium. It monitors and analyzes users' emotional states and attention levels to provide personalized task planning, reminders and companionship. Additionally, it integrates therapeutic doodling activities and a reward system to alleviate stress and boost motivation. Our research demonstrates that LittleToDo effectively mitigates procrastination while showing positive effects on stress reduction and self-efficacy enhancement. This study contributes an innovative intervention pathway to address academic procrastination among adolescents.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**; **Empirical studies in HCI**; **Interaction devices**; **Empirical studies in interaction design**.

Keywords

AI Agent, Academic Procrastination, Affective Computing, Expressive Art Therapy

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1 Introduction

Academic procrastination, a widespread habit among students of unnecessarily delaying academic tasks, leads to increased stress, poor performance, and reduced well-being [16]. This issue is particularly prevalent in Asian countries like Singapore, China, and Japan due to high academic expectations and intense competition [11, 19, 41, 47–49]. Asian adolescents face greater academic pressure and score higher on procrastination scales [16], contributing to psychological and behavioral issues such as depression, anxiety, and even suicide [11]. Despite its severity, current educational practices lack effective solutions, with harsh interventions often exacerbating the problem [55]. Academic procrastination is fundamentally an emotional regulation issue, misinterpreted as laziness or lack of self-discipline, leading to a cycle of chronic procrastination and psychological distress [23, 33].

Multiple factors, including heavy coursework, lack of confidence, insufficient motivation, anxiety, and some clinical conditions such as focus issues and innate Attention-Deficit / Hyperactivity Disorder (ADHD) [31], can contribute to academic procrastination [7, 16]. Traditional interventions, like psychological counseling and Cognitive Behavioral Therapy (CBT), have shown promise but require persistent adherence and can be costly [14, 15, 54]. Additionally, task management apps, while helpful, lack sufficient reinforcement mechanisms and can increase anxiety [1, 2]. Home-based interventions are often ineffective due to parents' lack of professional training [47].

This study introduces LittleToDo, an on-desk device using AI technologies to record, analyze, and engage in conversations with users. It assesses emotional states, analyzes stress and procrastination levels, and provides personalized study plans and guidance. The AI agent interacts through speech, and responses are printed via a mini printer also used for art-based therapeutic interactions to support emotion regulation and stress relief [47]. The aim is to provide an innovative, comprehensive intervention for adolescent academic procrastination. The contributions of this research include the following:

1. We designed a compact on-desk device driven by large language models (LLMs) AI agent, primarily for intervening in adolescent academic procrastination.
2. We explored different intervention methods for adolescent procrastination and integrated them into our device.

3. We designed an LLM-based affective computing strategy that utilizes multimodal data, including natural language dialogue records and facial expression analyses, as structured input to enhance user interactions with the LLM by adapting to their emotional states.

4. We evaluated the effectiveness and user experience of the device through user study.

2 RELATED WORK

2.1 Factors Influencing Academic Procrastination

Academic procrastination among adolescents is a multifaceted issue influenced by various interconnected factors, manifesting both before and during task execution. A primary factor is self-regulation failure, where students struggle with decision-making and conflict resolution [39]. This challenge is exacerbated in current educational systems focused on examinations, leading to increased workloads that surpass adolescents' management capabilities [18, 26, 51]. Anxiety also plays a crucial role, as adolescents often avoid tasks due to fear of failure when faced with overwhelming workload and difficulty [17, 44]. This fear generates anxiety, leading to a cycle where setbacks are perceived as intolerable, prompting procrastination. Additionally, loneliness can diminish academic interest and motivation [6, 24].

2.2 Approaches to Alleviate Academic Procrastination

Effective strategies to combat procrastination include task planning and management. By establishing specific schedules, tasks can be handled systematically and workloads balanced [5]. Traditional methods, such as notes or memos, remain popular despite their monotony. Modern mobile applications offer engaging features like gamification, making tasks more enjoyable and fostering long-term learning motivation [46]. For instance, Forest uses a 'tree-growing' reward system to minimize phone usage and incentivize focus [1]. However, many apps lack flexibility, failing to adapt to users' mental states and energy levels, and may inadvertently promote phone usage [3]. Ideally, task planning tools should be personalized, adjusted based on users' emotional states, task difficulty, and progress.

Psychological intervention is crucial in addressing academic procrastination. Emotional support from teachers, parents, or peers provides positive reinforcement, benefiting academic performance [8, 22, 52]. Positive peer relationships enhance learning motivation and engagement [42]. Addressing loneliness, a key procrastination factor, requires emotional support and companionship [38]. AI technologies, such as chatbots, offer companionship and can mitigate loneliness [29, 53]. Individuals tend to demonstrate honesty and self-disclosure with conversational AI [32]. Advances in affective computing and LLMs enable emotionally intelligent AI companions that provide empathetic and reliable psychological support [12, 21], encouraging self-reinforcement to reduce procrastination [9, 21]. However, maintaining long-term engagement with chatbots remains challenging, traditional AI chatbots and even LLM-based chatbots have limitations in understanding emotions through language interactions [10].

Art-based interventions are effective in managing stress and anxiety, fostering calmness [30]. Simple artistic activities, like drawing, clay modeling, and crafting, reduce anxiety symptoms and attention deficits [27]. Mandala coloring, in particular, creates psychological spaces that reduce defensive barriers and enable safe self-expression [37, 43]. These non-verbal techniques allow participants to externalize emotions and internal conflicts [28]. While generative AI assistance can mitigate variations in perceptual and creative abilities [20, 40], research on the specific relationship between artistic expression and academic procrastination is insufficient, and accessible therapeutic tools are lacking.

In summary, academic procrastination among adolescents is driven by multiple factors and requires comprehensive intervention approaches. Existing studies often address partial symptoms, leaving a gap in developing a holistic support system that addresses task management, emotional issues, companionship, and communication simultaneously.

3 FORMATIVE STUDY

To investigate the underlying causes and potential interventions for academic procrastination among adolescents, we conducted a questionnaire survey with 50 adolescents (25 MALES, 25 FEMALES; MEAN AGE=13.06, SD=±1.26), followed by in-depth online interviews with 27 self-identified procrastinators. Of the interviewed participants, over 30% exhibited severe procrastination behaviors, while the remainder demonstrated mild to moderate tendencies. Analysis of the data revealed that most adolescents had attempted self-management strategies, such as creating study plans or using time management tools, but few were able to sustain these efforts. Four key factors contributing to procrastination were identified: (1) ineffective time management and task allocation, (2) distractions, (3) negative emotional states, and (4) lack of supportive companionship.

Building on these findings, our objective is to design an effective intervention device that actively understands and engages users, provides tailored task suggestions, and facilitates simple stress-relief activities. The proposed device should be compact enough to seamlessly integrate into learning environments, function independently of a smartphone, and incorporate AI technologies—particularly emotional intelligence and affective interaction strategies.

4 LITTLETODO: SYSTEM DESIGN AND DEVELOPMENT

A cross-disciplinary team comprising members with backgrounds in interaction design and computer science collaborated to develop LittleToDo, which achieves deep hardware-software integration to understand users' emotional states and intervene when needed with personalized task planning suggestions and consistent companionship.

4.1 Hardware and Software

The hardware of LittleToDo consists of an Orange Pi Zero 3 (quad-core Cortex-A53 processor, 1GB RAM) and integrates key components including a thermal printer, USB wide-angle camera, and microphone. All hardware components are housed within a compact

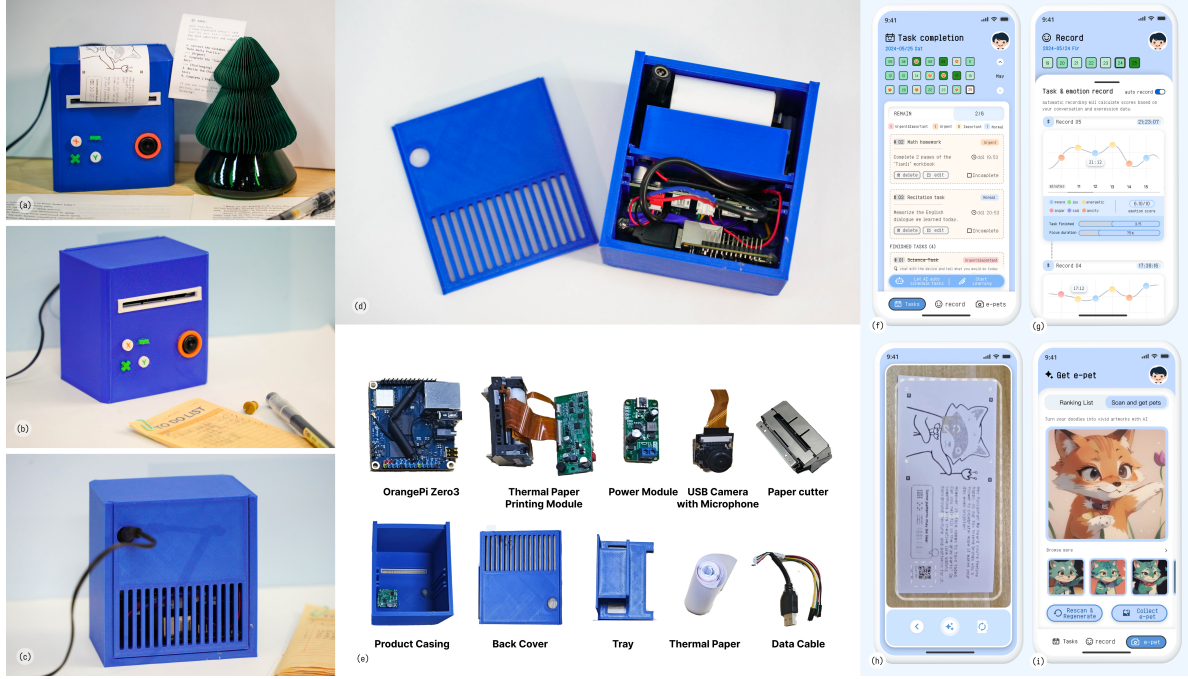


Figure 1: Appearance, hardware and APP interface design of LittleToDo.

3D-printed casing measuring 95mm × 65mm × 105mm, designed for convenient placement on a desk, as shown in (Figure 1(a) to (e)). The system utilizes client-server architecture, with the core application running on the Orange Pi hardware device, while server-side API interfaces provide computational support for AI. Additionally, (Figure 1(f) to (i)) illustrates the mobile application we developed, which facilitates data visualization and offers supplementary interaction features, enabling users to track learning progress and manage tasks flexibly.

Users can interact with the device directly through natural voice input and gesture controls, without the need for a mobile phone. The device provides feedback by printing paper notes via its miniature printer. During voice-based chat interactions, the device processes the information captured and prints responses; Built-in quick functions such as activating focus mode or postponing tasks are offered through gesture control.

4.2 Affective Computing — Driven AI Agent

To establish our device as an empathetic learning companion, we focused on creating emotional connection between technology and users [35]. Previous research indicates that inputting GenAI systems with data obtained during Facial Emotion Recognition (FER) stages facilitates the generation of contextually relevant and emotionally connected responses [13].

During user-device interactions, both linguistic and visual data are captured, processed, and analyzed to create digitized emotion and behavioral records for each user. The flowchart is shown in (Figure 2(a)). Specifically, the integrated HD camera captures images of users at 5-second intervals during study sessions. These images are sent to the multimodal Llama 3.2 Vision 11B model [4]

to analyze learning states and detect distracting behaviors, such as phone usage or daydreaming. Additionally, facial images are extracted and processed separately. We extended and trained our emotion recognition model based on the FER2013 dataset, while implementing Google MediaPipe for facial landmark analysis, enabling the device to assess users' emotional states and fatigue levels based on facial expressions. Furthermore, during user-device conversations, a Qwen2.5-3b-based agent [36] analyzes and classifies users' emotions based on their conversation rate per minute. To optimize the flow efficiency, we implemented Dify [25] to run this automation system on our servers, where all emotional data is weighted, structured, and recorded in the database.

When the device needs to respond, another workflow powered by Qwen2.5-72B triggered. Both daily and recent (last hour) emotional and behavioral data records are incorporated into the prompt instructions, as shown in (Figure 2(b)). This workflow executes different actions based on the user's current emotional state and task schedule, such as offering encouragement or adjusting task sequences and scheduling breaks accordingly.

4.3 Task Management

When aiding their task planning, the system dynamically adjusts priorities based on the difficulty, urgency and users' overall states. Once everything is set, it prints a customized task list and provides reminders in case of prolonged inactivity or distraction. Notably, when negative mood is detected, it prioritizes easier tasks and prints a new schedule.

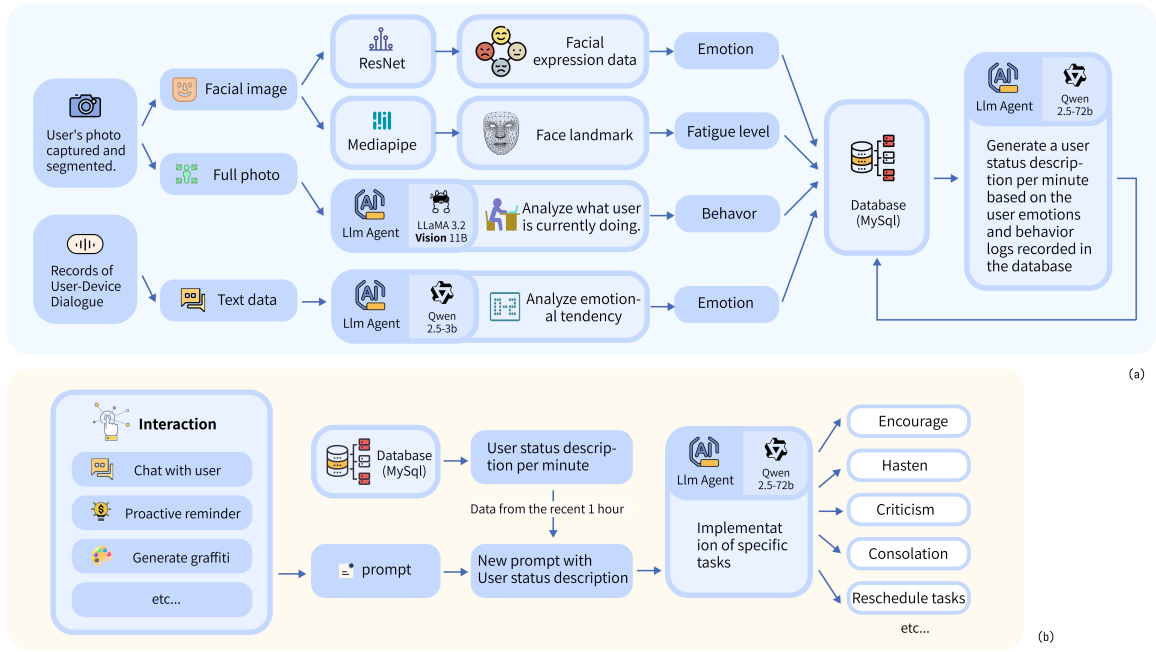


Figure 2: (a) Workflow for emotional data processing and recording, (b) Workflows that interact with sentimental data.

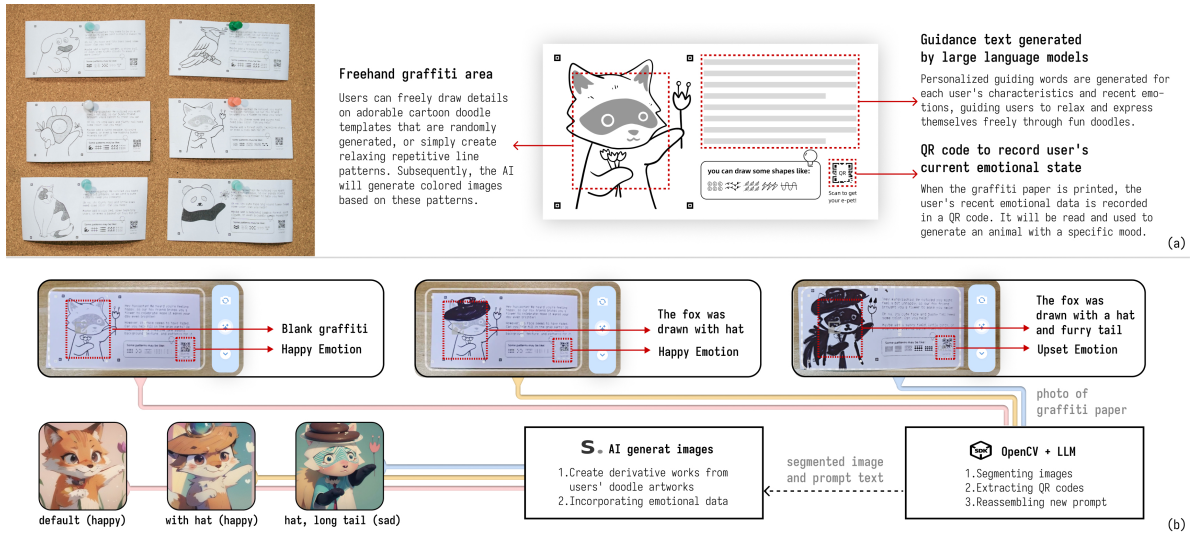


Figure 3: (a) Random scribble cards, (b) Process for AI generate images based on user-created content and user emotions.

4.4 Stress Relief and Relaxation

As a reliable learning companion, the device proactively prints encouraging or amusing messages to uplift the user's mood when prolonged periods of low mood are detected. To motivate sustained task performance, the device incorporates a reward mechanism inspired by art therapy principles. Upon completing a specified number of tasks, users receive a random doodle card with animal illustration for creative expression, shown in (Figure 3(a)), with

each card containing a QR code that records users' emotional state of that day. When scanning the completed artwork in our mobile application, AI generates unique effects based on the recorded emotional state, the processing flow is shown in (Figure 3(b)). Users can share or exchange their doodle artworks with friends, further enhancing a sense of engagement.

5 USER STUDY

5.1 Participants and Hypotheses

To validate our hypotheses and evaluate the effectiveness and user experience of LittleToDo in practical applications, we recruited 12 adolescents who self-identified as having mild procrastination tendencies from our previous questionnaire study. The participant group consisted of 6 males and 6 females, aged 12 to 14 years (MEAN=12.92, SD = ± 0.67). Our preliminary assessment confirmed that all participants were proficient in using various digital devices such as smartphones and tablets, with two-thirds reporting prior experience with AI agents. The experiment lasted for two weeks, during which all participants were thoroughly briefed on the study's objectives and procedures. Informed consent was obtained through formal documentation.

The experiment was designed to test three primary hypotheses: Hypothesis 1: LittleToDo demonstrates significant effectiveness in helping adolescents mitigate academic procrastination behaviors. Hypothesis 2: LittleToDo effectively assists students in alleviating anxiety and reducing stress. Hypothesis 3: LittleToDo effectively enhances adolescents' academic self-efficacy.

5.2 Procedure

Before the experiment, we conducted a baseline assessment for all participants utilizing a comprehensive evaluation scale developed based on expert recommendations. The 48-items scale encompassed three dimensions: level of academic procrastination, stress, and self-efficacy. We modified the original 18 questions in the first section of the Procrastination Assessment Scale-Students (Solomon and Rothblum, 1984) [45] to better suit adolescent students. Stress level was evaluated using the Positive and Negative Affect Schedule (PANAS) 5-point scale [50], which evaluates 20 positive and negative emotions. The self-efficacy component was developed based on the Motivated Strategies for Learning Questionnaire (MSLQ) by Pintrich & De Groot (1990) [34]. This section includes 10 questions specifically designed to evaluate adolescents' confidence and beliefs in their ability to complete specific tasks. To ensure each participant fully understood each scale items, we read out the content and clarified any questions they had.



Figure 4: Teenagers use and interact with LittleToDo.

Next, participants underwent a one-hour training session where they were instructed on how to use our device and its mobile application. They were required to use LittleToDo to organize their after-school learning tasks over the following two weeks. Participants were instructed to keep the device activated and interact with it during their after-school study sessions (Figure 4). This enabled

the analysis of their emotional states and task completion processes, as well as the delivery of tailored interventions. Additionally, we collected daily after-school workload data from each participant's head teacher to ensure that their academic task volumes during the experimental period remained consistent with pre-experiment levels. This helped maintain uniform study conditions and controlled for potential workload-related variables that might influence the results. Two weeks later, we conducted a second round of assessment using comprehensive scale and investigated deeper insights into user experiences through semi-structured interviews. We also conducted interviews with the participants' teachers and parents to evaluate their learning states and procrastination levels and to record their perspectives on and acceptance of the device.

6 RESULTS AND DISCUSSION

6.1 Hypotheses Evaluation

Following the experiment and the collection of all user interview data and scale results, we performed a weighted and quantitative analysis of each scale section. Figure 5 depicts the data distribution for each dimension. In Table 1, we analyzed the differences between pre- and post-experiment data using the Wilcoxon signed-rank test. The results reveal significant differences across all three dimensions before and after the two-week usage of the LittleToDo device. Regarding H1, pre-experiment procrastination assessment scores were higher than post-experiment scores, demonstrating LittleToDo's significant effectiveness in helping adolescents mitigate academic procrastination behaviors. For H2, post-experiment positive emotion scores increased while negative emotion scores decreased, confirming the device's supportive role in alleviating academic stress. As for H3, the increase in academic self-efficacy assessment scores after device usage indicated enhanced confidence and self-discipline in completing homework tasks.

As shown in Table 2, with participants' consent, we exported and analyzed device logs using a related sample t-test, including the proportion of focus time to total time and the completion rates of learning tasks, which were automatically recorded by AI models. Compared to the first day of the experiment, users demonstrated a significant increase in attention duration and task completion rates. This improvement closely aligned with their self-assessment results, indicating a reduction in procrastination and a boost in learning motivation following the intervention.

6.2 Discussion

6.2.1 Effectiveness of LittleToDo. During interviews, 9 students reported reduced homework completion times, attributing this to the device's reminders, which effectively motivated them and enhanced their time management skills. Furthermore, more than half of the participants mentioned decreased anxiety when initiating new tasks and improvements in task execution abilities after one week of use. Feedback from parents highlighted increased proactivity in their children's approach to tasks, with timely homework completion allowing for more free time, family interactions, and participation in extracurricular activities. Moreover, a head teacher reported an increase in the number of assignments submitted before deadlines during the two-week period. Although three participants still failed to complete all their after-school learning tasks on time,

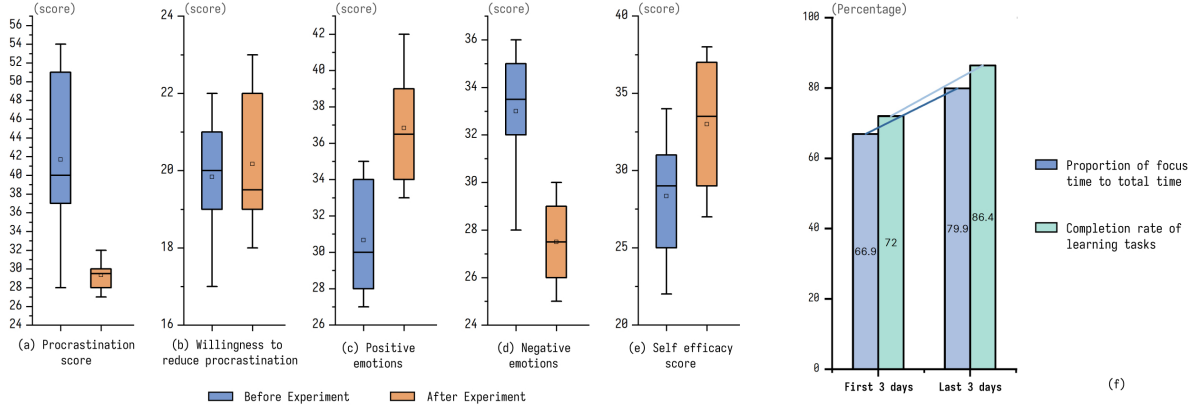


Figure 5: (a) Distribution of participants' scale assessment data before and after the trial, (b) Distribution of system log data at the beginning and end of the trial.

Table 1: The statistical metrics on participants' performance

Metrics	Distribution M(P25, P75)		Statistics		Hypotheses
	Before	After	Z	p (* p<0.05 ** p<0.01)	
Procrastination	40.0(37.0,51.0)	29.5(28.0,30.0)	2.839	0.005**	H1 accepted
Willingness to reduce procrastination	20.0(19.0,21.0)	19.5(19.0,22.0)	0.563	0.574	
Positive emotions	30.0(28.0,34.0)	36.5(34.0,39.0)	3.105	0.002**	H2 accepted
Negative emotions	33.5(32.0,35.0)	27.5(26.0,29.0)	2.839	0.005**	
Self-efficacy	29.0(25.0,31.0)	33.5(29.0,37.0)	3.076	0.002**	H3 accepted

Table 2: Statistics of logs exported from the device for participants

Metrics	Distribution (M ± SD)		Statistics	
	Before	After	Z	p (* p<0.05 ** p<0.01)
Proportion of focus time to total time	66.92±13.56	79.92±9.42	-3.305	0.007**
Completion of learning tasks	72.00±10.97	86.42±13.23	-14.42	-5.595

they nonetheless exhibited varying degrees of improvement in their task performance.

6.2.2 User Experience. User feedback reflects that participants found the companionship provided by LittleToDo and its paper note communication method engaging. The majority reported feeling understood and cared for by LittleToDo device, which helped alleviate their emotional burden and academic pressure. As one 13-year-old male participant remarked: “When I delayed starting my studies due to video games, this box seemed to perceive my executive functioning issue, subsequently encouraging me to complete my learning tasks before playing video games.” Similarly, another 12-year-old female participant shared: “LittleToDo feels like a close friend. When I was stuck on a math problem, it gently reminded me to take a break and seek new inspiration instead of staying frustrated.” Besides, the device’s doodling functionality introduced a fun and relaxing element to studying time. According to one teacher, students compared their animal doodle collections with the animals’ expressions helping them recognize unconscious

emotional fluctuations. This interaction not only promoted mutual encouragement among classmates but also fostered self-regulation.

6.3 Limitations and Future Work

Our research on an intervention device for adolescent academic procrastination shows promise, but limitations in experimental duration and sample size necessitate further validation. While the AI models were effective for most participants, variations may exist across age groups and personality types. Future work will refine emotion evaluation methods and develop personalized strategies to enhance the device’s effectiveness across diverse populations.

7 CONCLUSION

This research investigated academic procrastination among adolescents and developed LittleToDo, a compact on-desk tool addressing this issue. Through formative investigation, we identified specific challenges faced by adolescents and established clear design objectives. LittleToDo adopts an LLM-based AI agent as its core component and implements multiple pathways for procrastination

intervention, including task planning and management, textual reminders, and art-based therapeutic. Our research demonstrates significant improvements in three key areas: reduced procrastination behavior, lowered stress levels, and enhanced self-efficacy. Users demonstrated willingness to engage with the device for task organization and emotional insights, while teachers and parents reported positive changes in teenagers' attitudes toward task execution. By developing and validating this tool, our research contributes an exploratory solution to adolescent academic procrastination, representing a valuable attempt to combine affective computing and AI agent applications in educational contexts.

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A Assessment scale

Comprehensive Evaluation of Academic Procrastination in Adolescents

PART 01 : Procrastination Assessment Scale for Students

Activities :

- I. Finishing homework
- II. Studying for exams
- III. Keeping up with weekly reading assignments
- IV. Academic administrative tasks: Filling out forms, registering for classes, getting id card
- V. Attendance tasks: Meeting with teachers, asking for help with learning issues, reporting on the progress of studies
- VI. School activities in general.

Questions :

For each of the activities above, please rate the degree to which you delay or procrastinate.

1. To what degree do you procrastinate on this task?

1. Never Procrastinate 2. Almost Never 3. Sometimes 4. Nearly Always 5. Always Procrastinate

2. To what degree is procrastination on this task a problem for you?

1. Never Procrastinate 2. Almost Never 3. Sometimes 4. Nearly Always 5. Always Procrastinate

3. To what extent do you want to decrease your tendency to procrastinate on this task?

1. Do Not Want to Decrease
2. Between 1 and 3
3. Somewhat Want to Decrease
4. Between 3 and 5
5. Definitely

PART 02 : Positive and Negative Affect Schedule

Indicate the extent you have felt this way over the past week.

(1. Very slightly or not at all 2. A little 3. Moderately 4. Quite a bit 5. Extremely)

- | | |
|---------------|-----------|
| 1. Interested | 1 2 3 4 5 |
| 2. Distressed | 1 2 3 4 5 |
| 3. Excited | 1 2 3 4 5 |
| 4. Upset | 1 2 3 4 5 |
| 5. Strong | 1 2 3 4 5 |
| 6. Guilty | 1 2 3 4 5 |

- | | |
|-----------------|-----------|
| 7. Scared | 1 2 3 4 5 |
| 8. Hostile | 1 2 3 4 5 |
| 9. Enthusiastic | 1 2 3 4 5 |
| 10. Proud | 1 2 3 4 5 |
| 11. Irritable | 1 2 3 4 5 |
| 12. Alert | 1 2 3 4 5 |
| 13. Ashamed | 1 2 3 4 5 |
| 14. Inspired | 1 2 3 4 5 |
| 15. Nervous | 1 2 3 4 5 |
| 16. Determined | 1 2 3 4 5 |
| 17. Attentive | 1 2 3 4 5 |
| 18. Jittery | 1 2 3 4 5 |
| 19. Active | 1 2 3 4 5 |
| 20. Afraid | 1 2 3 4 5 |

PART 03 : Evaluation Of Self-Efficacy For Studying Tasks

Rate each following item on an “1” to “5” scale.

(1. Completely not applicable; 2. Somewhat not applicable; 3. Indeterminate; 4. Somewhat applicable; 5. Completely applicable)

1. I believe that I have the ability to complete my study tasks and homework well.
2. I think I have the ability to solve problems encountered in study tasks and homework after class.
3. Compared with other students in my class, my ability to complete learning tasks is stronger.
4. I like to choose challenging learning tasks.
5. When work is difficult, I either give up or learn only the easy parts.
6. I often choose learning tasks that are difficult but from which I can learn, even if it requires more effort.
7. I always mark out key parts in books or notebooks to help with my study.
8. Even if the study material is boring, I will stick to it until I finish it.
9. When I do my homework, I always try to remember what the teacher said in class so that I can do my homework well.
10. Even if the teacher did not ask me, I would consciously do the exercises at the end of each chapter in the book to test my mastery of knowledge.

Figure 6: Appendix assessment scale.